Problem Statement

Real-Time Human Detection on Railway Line Using Computer Vision and Deep Learning

Railway transportation is one of the most widely used and critical modes of transport. However, railway accidents caused by unauthorized human presence on railway tracks remain a major safety concern. Trespassing, accidental crossing, maintenance worker exposure, and suicide attempts often lead to severe injuries, fatalities, delays in train operations, and damage to railway infrastructure. Traditional monitoring systems mainly depend on manual surveillance through security personnel or CCTV operators, which becomes inefficient for continuous real-time monitoring over large railway networks.

The increasing availability of surveillance cameras and advancements in Artificial Intelligence, Computer Vision, and Deep Learning provide an opportunity to develop automated railway safety systems capable of detecting humans on railway tracks in real time. Such systems can assist railway authorities by continuously monitoring railway environments and generating instant alerts whenever a person is detected within dangerous zones near or on the railway line.

The proposed system aims to develop a **Real-Time Human Detection on Railway Line** solution using deep learning-based object detection and/or segmentation techniques. The system processes live video streams or CCTV footage to identify humans present on or dangerously close to railway tracks. Once a human is detected, the system can trigger alerts, warnings, or notifications to railway control systems, station authorities, or train operators for immediate action.

The major challenge in this problem is achieving high detection accuracy under real-world conditions such as varying lighting environments, rain, fog, shadows, occlusions, crowded scenes, camera vibrations, and long-distance visibility. The system must also minimize false positives caused by objects such as poles, animals, bags, or railway equipment while maintaining low-latency performance suitable for real-time deployment.

To address these challenges, the system will utilize advanced computer vision models such as YOLO-based object detection or segmentation architectures trained on railway-specific datasets. The model should be capable of accurately detecting humans across different track conditions, camera angles, and environmental scenarios. The final solution should support efficient inference speed, scalability, and practical deployment in railway surveillance infrastructure.

The objective of this project is to enhance railway safety by reducing accidents and enabling intelligent automated monitoring through AI-powered real-time human detection systems.